Name: Hamizah Aziim Bhikan

Roll No:DS24105

Title : Analyzing the Impact of Global Events on Gold Prices Using Machine Learning Models

1-Abstract

Gold prices exhibit significant volatility due to their strong sensitivity to economic uncertainty, inflation, currency fluctuations, financial crises, and sudden global events. During periods of instability, investors tend to shift capital from risky assets toward gold, leading to sharp and unpredictable price movements. Traditional forecasting techniques often fail to capture these rapid changes because they are limited in modeling complex and non-linear market behavior. This study examines the impact of major global events on gold price movements using historical gold price data from 2000 to 2025, a period that includes multiple economic cycles, financial crises, geopolitical conflicts, and global disruptions such as the COVID-19 pandemic and the Russia–Ukraine conflict. To analyze and predict gold prices, machine learning models was employed, Linear Regression, Random Forest, XGBoost, and Long Short-Term Memory (LSTM). The performance of these models was evaluated to assess their ability to handle volatility and sudden market shocks. The results indicate that the LSTM model performs traditional statistical and other machine learning models due to its strength in capturing long-term temporal dependencies and complex non-linear patterns in time-series data. The findings confirm that gold price volatility is closely linked to global economic and geopolitical events, and that advanced machine learning approaches, particularly deep learning models, provide more accurate and reliable predictions. This study demonstrates that integrating global event-driven market behavior with modern machine learning techniques significantly enhances gold price forecasting accuracy.

2-Keywords:

Gold prices, Machine Learning, Global events, Financial forecasting, Event-driven models, Linear Regression, Random Forest,LSTM , Time-series analysis

3-Introduction

Gold is one of the most trusted investment assets in financial markets around the world. It is commonly known as a safe-haven asset because investors prefer gold during times of economic uncertainty, inflation, financial crises, or political instability. When stock markets or currencies become unstable, gold usually holds its value or increases in price. Because of this stability, gold is widely used by individual investors, financial institutions, and central banks as a protection against risk.

In the past, gold price prediction was mainly done using traditional statistical and economic methods. These methods included trend analysis, moving averages, inflation indicators, exchange rate analysis, and time-series models such as ARIMA. Most of these approaches depended only on past price data and assumed that market behavior would remain stable. Although these techniques worked reasonably well under normal conditions, they often failed when sudden changes occurred in the market.

Global events have played a major role in increasing gold price volatility. Events such as the COVID-19 pandemic, the Russia–Ukraine conflict, global financial crises, and political tensions have caused unexpected changes in gold prices. During the COVID-19 pandemic, fear and uncertainty pushed investors toward gold, resulting in sharp price movements. Similarly, wars and political conflicts disturbed global markets and increased demand for gold. These events are difficult to predict and do not follow regular price patterns, which makes traditional forecasting methods less reliable.

To address these challenges, machine learning techniques have gained attention in financial analysis. Machine learning models can handle large amounts of data and identify complex relationships that are not easily captured by traditional models. By using historical gold prices along with information related to global events and market uncertainty, machine learning models can better adapt to changing market conditions. This makes them suitable for studying gold price behavior during uncertain periods.

The main objective of this paper is to explore how machine learning methods can be used to analyze the effect of global events on gold price movements. The study focuses on understanding how combining historical price data with global event information can improve prediction accuracy. This research provides a basic framework that can support future practical studies and the development of machine learning-based gold price prediction models.

4-Objectives of the Study

1. To analyze historical gold price movements and identify major influencing global events.
2. To study the relationship between global crises and gold price volatility.
3. To build machine learning models capable of predicting gold price fluctuations.
4. To compare the performance of different machine learning algorithms.
5. To determine which events have the strongest impact on gold price trends.
6. To assist investors and policymakers by providing insights based on event-driven analysis.

5. Literature Review

5.1 Determinants of Gold Prices

Gold has always been considered a safe and stable investment, especially during economic and financial uncertainty. Many researchers have studied the factors that influence gold prices. Inflation is one of the most important factors. According to Singh and Bhanawat (1), when inflation rises, investors prefer gold to protect their purchasing power. Hussain and Rehman (10) also explain that inflation and currency depreciation together increase gold prices.

Exchange rate movements also affect gold prices. Kumar and Patel (3) find that gold prices usually move opposite to the U.S. Dollar Index. When the dollar weakens, gold becomes more attractive for investors. Jeon and Kim (2) show that during periods of economic instability, demand for gold increases due to its safe-haven nature.

Investor behavior and market fear also play an important role. Chen and Wu (6) show that fear indicators like the VIX index have a positive relationship with gold prices. Studies by Alqahtani (4) and

Qureshi and Hassan (7) show that geopolitical tensions and political uncertainty increase gold price volatility. Sahu (9) explains that during economic slowdowns, investors shift their funds towards gold.

Central banks also influence gold prices. Smith and Carter (11) report that during global crises, central banks increase gold reserves, which impacts international gold prices. Overall, the literature confirms that gold prices depend on economic, psychological, and institutional factors.

5.2 Impact of Global Events on Gold Price Fluctuations

Global events act as sudden shocks that strongly affect gold prices. Many studies focus on how pandemics, wars, and economic crises influence gold markets. During the COVID-19 pandemic, gold prices increased sharply as investors avoided risky assets. Jeon and Kim (2) and Sharma (5) observe that gold performed strongly during the pandemic due to high uncertainty.

Economic recessions further strengthen gold's role as a safe asset. Kaur (8) and Sahu (9) show that gold prices either remain stable or increase during recessions. Geopolitical conflicts also have an immediate impact on gold prices. Smith and Carter (11) analyze the Russia–Ukraine conflict and report price spikes during the early stages of the war.

Political events and policy announcements also affect gold prices. Baral and Pokharel (12) find that elections and political instability increase commodity price volatility. Dahiya and Verma (14) show that monetary policy announcements by central banks, especially the U.S. Federal Reserve, cause quick reactions in gold markets.

These studies clearly show that global events cause abnormal movements in gold prices, which cannot be captured properly by traditional models.

5.3 Machine Learning Models for Gold Price Prediction

Due to increasing complexity in financial markets, researchers have started using machine learning techniques for gold price prediction. Patel and Mehta (15) show that models such as Linear Regression, Decision Trees, and Random Forest perform better than traditional econometric models. Kumar and Tiwari (16) compare different ML models and find that non-linear algorithms handle gold price volatility more effectively.

Ensemble models have gained popularity due to their higher accuracy. Sharma and Singh (18) report that XGBoost performs well in forecasting gold prices. Thomas and George (17) also show that tree-based models give better results than simple regression models.

Deep learning techniques further improve prediction accuracy. Gupta and Roy (19) demonstrate that neural networks perform well when trained on large datasets. Mani and Rajan (20) find that LSTM models are especially suitable for time-series forecasting because they capture long-term dependencies. Other studies also confirm that ML models generally outperform traditional statistical methods (21, 22).

However, most ML studies rely mainly on historical price data and technical indicators.

5.4 Event-Based and Hybrid Machine Learning Models

Recent studies focus on combining machine learning models with event-based information. Martin and Lopez (23) propose an event-driven ML framework that includes political events and economic shocks, resulting in better forecasting accuracy.

Sentiment-based approaches are also widely used. Ahmed and Farooq (24) show that news and social-media sentiment improve gold price prediction during crisis periods. Hybrid models that combine statistical and ML techniques are another important research direction. Goyal et al. (25) and Yadav (26) find that hybrid ARIMA-ML models capture both linear and non-linear patterns in gold prices.

Patel et al. (27) compare ARIMA, LSTM, and hybrid models and conclude that hybrid models perform better. NLP-based approaches are also emerging. Wei and Huang (29) and Singh and Rao (30) show that text mining and event extraction help in predicting sudden gold price changes.

Despite better performance, these models face challenges such as limited event data and difficulty in measuring event intensity.

5.5 Research Gap

Although many studies have analyzed gold price determinants and used ML models for forecasting, several gaps remain. First, many ML models focus only on historical prices and ignore global events (15, 16). Second, studies that include events usually focus on a single event type, such as pandemics (2, 5) or wars (11), rather than multiple events together.

Another major issue is interpretability. Advanced ML and hybrid models often work as black boxes, making it difficult to understand the impact of individual events (25, 27). Moreover, global sentiment data is still under-utilized, even though gold is traded worldwide (24, 29).

Therefore, this study aims to provide a **simple and structured event-integrated machine learning framework** that explains how global events can be combined with ML models for gold price prediction, making it easier for new researchers to understand and apply.

6.Research Methodology

6.1 Data Collection

The dataset used in this study consists of monthly gold price data collected from DataHub (Core Gold Prices). The data is available in CSV format and is expressed in USD per ounce. The study considers the time period from January 2000 to December 2025, with a monthly frequency, resulting in 312 observations.

The dataset contains two main attributes: Date, recorded in YYYY-MM format, and Gold Price, representing the monthly closing price of gold. This time period was selected because it covers several major global economic and geopolitical events such as the 2008 Global Financial Crisis, 2011

European Debt Crisis, 2020 COVID-19 pandemic, and the 2022 Russia–Ukraine war. Hence, the dataset is suitable for analyzing long-term gold price trends and event-driven price movements.

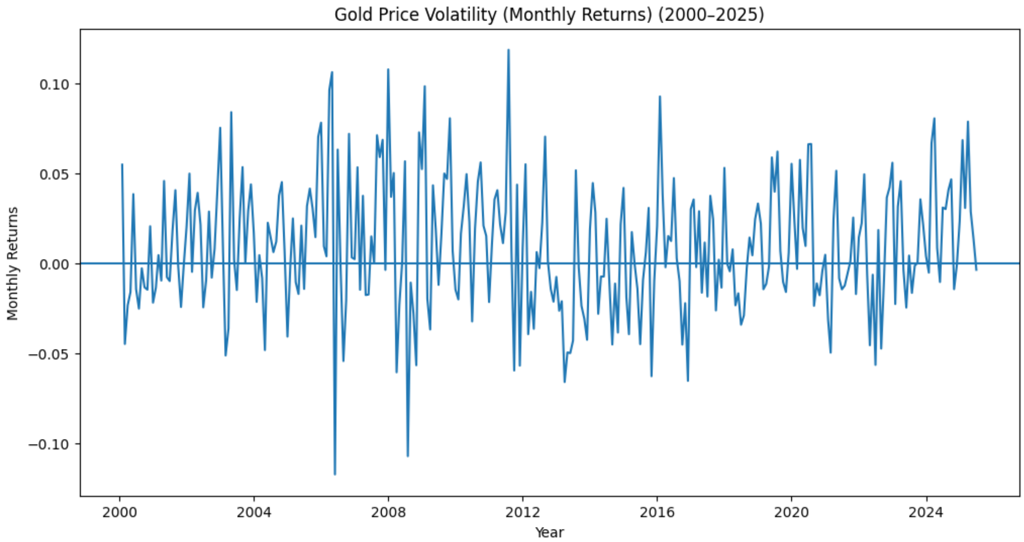
6.2 Data Pre-processing

Before analysis, the gold price data was pre-processed to make it suitable for further study. The date column was converted into a proper datetime format to ensure correct time-series ordering. The dataset was checked for missing values, and linear interpolation was used wherever required to maintain continuity.

Monthly returns were calculated to study price fluctuations during normal and crisis periods. Basic data cleaning was also performed by removing duplicate entries and verifying numeric values. The data was further checked for consistent monthly frequency. Finally, normalization was applied for machine learning models to improve training performance.

6.3 Exploratory Data Analysis (EDA)

1) To examine the relationship between global crises and gold price volatility, monthly gold price returns were analyzed over the period from 2000 to 2025.



Gold Price Volatility Based on Monthly Returns (2000–2025)

Figure illustrates the fluctuations in monthly gold price returns over the study period, highlighting periods of increased volatility. Sharp positive and negative spikes are observed during major global events such as the 2008 financial crisis, the COVID-19 pandemic, and recent geopolitical tensions. These extreme movements indicate heightened uncertainty in financial markets during crisis periods. The results confirm a strong relationship between global crises and increased gold price volatility, reinforcing gold's role as a safe-haven asset during times of economic and geopolitical instability.

2) To understand the long-term movement of gold prices, monthly gold price data from 2000 to 2025 was analyzed.

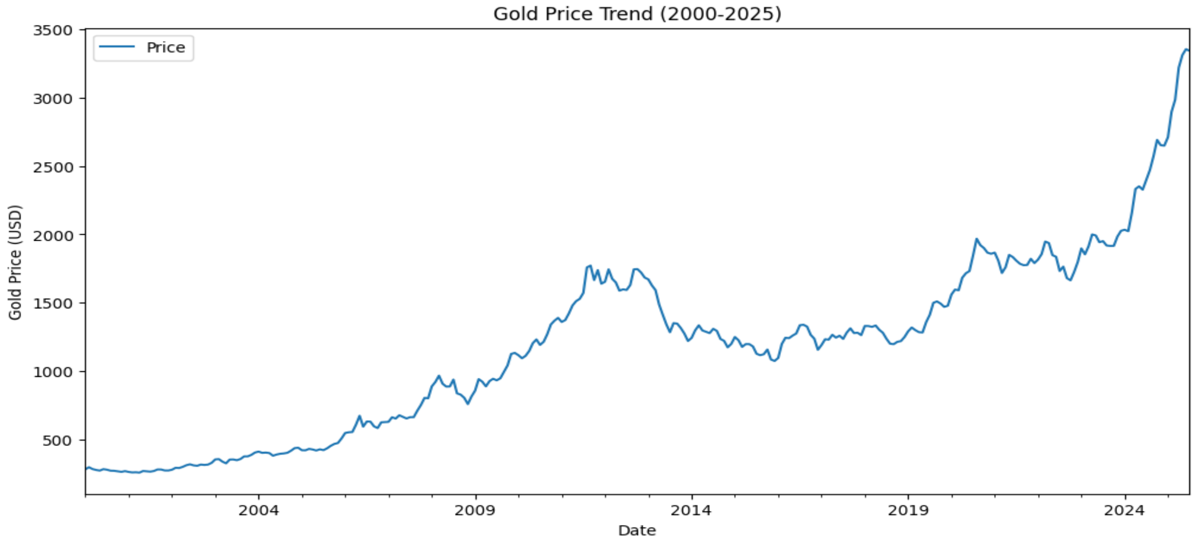


Figure illustrates the long-term trend in monthly gold prices over the study period. The graph shows a steady increase in gold prices with notable upward movements during periods of economic uncertainty, particularly around the 2008 global financial crisis and the COVID-19 pandemic. A temporary decline and stabilization can be observed between 2013 and 2018, followed by a sharp rise after 2020 due to increased global uncertainty and inflationary pressures. Overall, the trend indicates a strong long-term growth in gold prices, highlighting gold's role as a reliable store of value.

3) To examine the influence of major global economic and geopolitical events on gold prices, key crisis periods were highlighted on the gold price trend from 2000 to 2025.

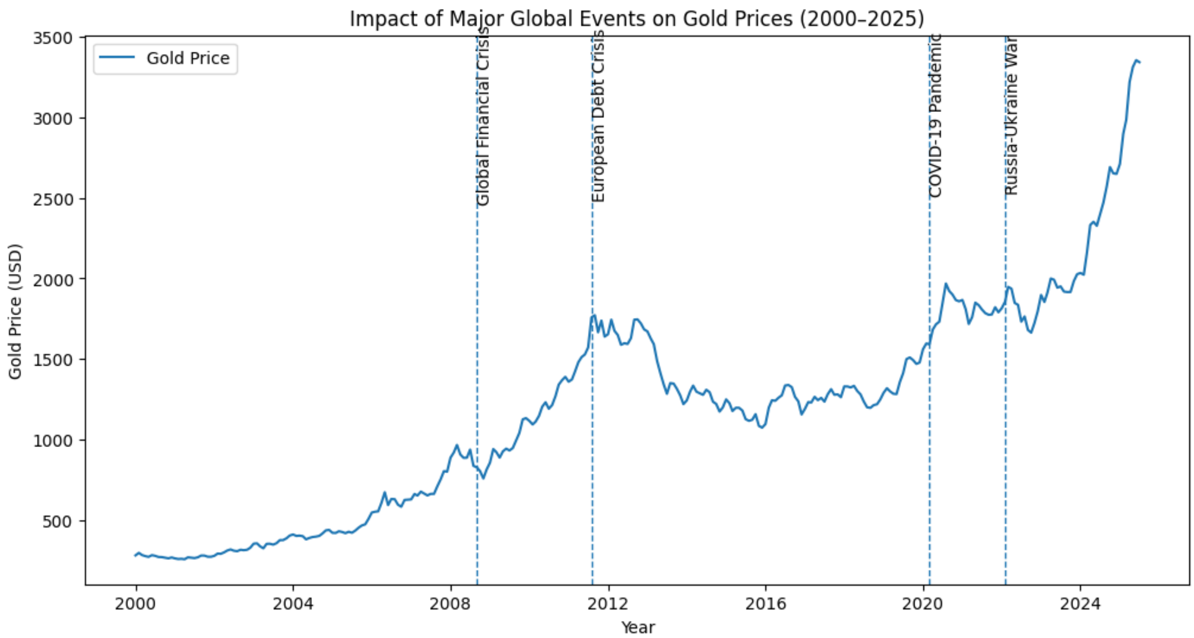


Figure demonstrates the response of gold prices to major global events, including the 2008 global financial crisis, the European debt crisis, the COVID-19 pandemic, and the Russia–Ukraine conflict. A noticeable upward movement in gold prices can be observed during and immediately after these events, reflecting increased investor demand for safe-haven assets. Periods of heightened uncertainty are associated with accelerated price growth and increased volatility. These observations confirm that global economic and geopolitical shocks have a significant impact on gold price movements.

6.4.Model Development

<u>Model Used:</u>

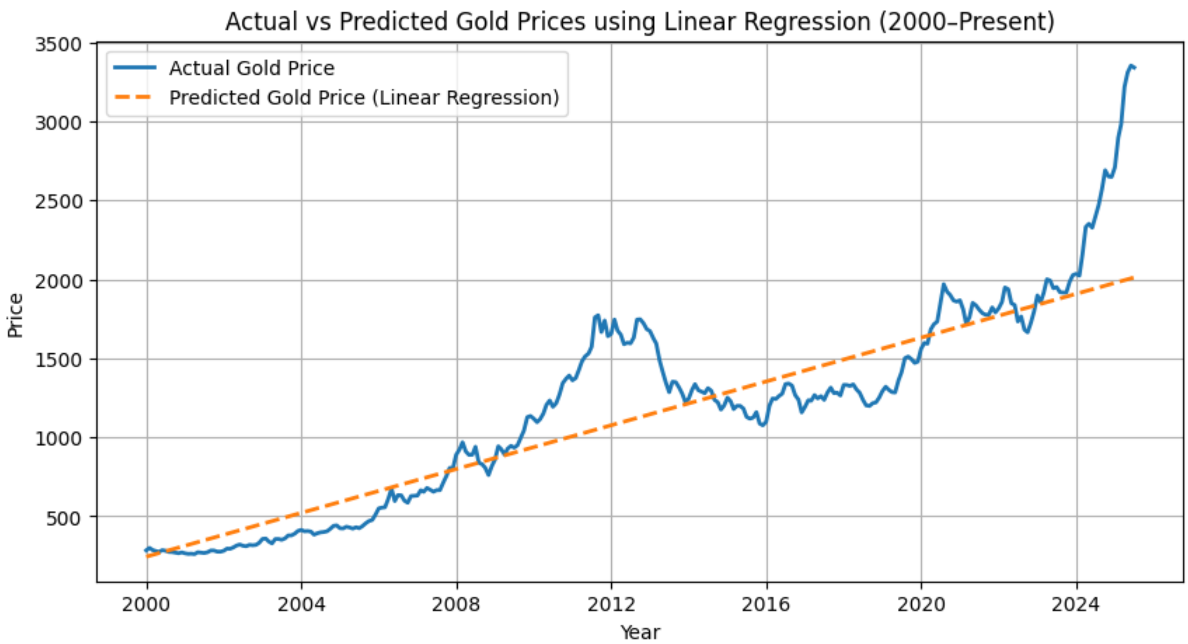
Linear Regression (LR): Linear Regression was used to analyze simple linear relationships between time and gold prices. It helps determine whether the gold price trend exhibits linear behavior.

Random Forest (RF): Random Forest is an ensemble-based model capable of capturing complex non-linear relationships. It is robust to volatility and helps model sudden fluctuations in gold prices.

XGBoost: XGBoost is a gradient boosting-based ensemble model that provides high predictive performance. It is effective in handling intricate patterns and sudden price spikes caused by global events.

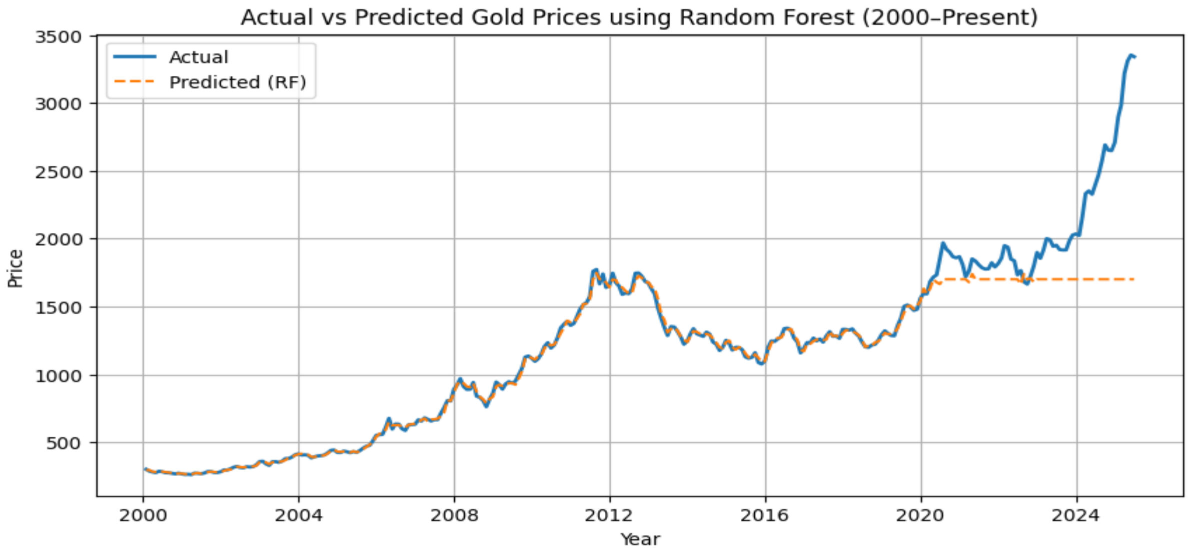
LSTM (Long Short-Term Memory): LSTM is a deep learning model designed for sequential data. It can learn long-term dependencies and complex temporal patterns, making it particularly suitable for predicting gold price movements during volatile periods.

1)Actual vs Predicted Gold Prices (Linear Regression)



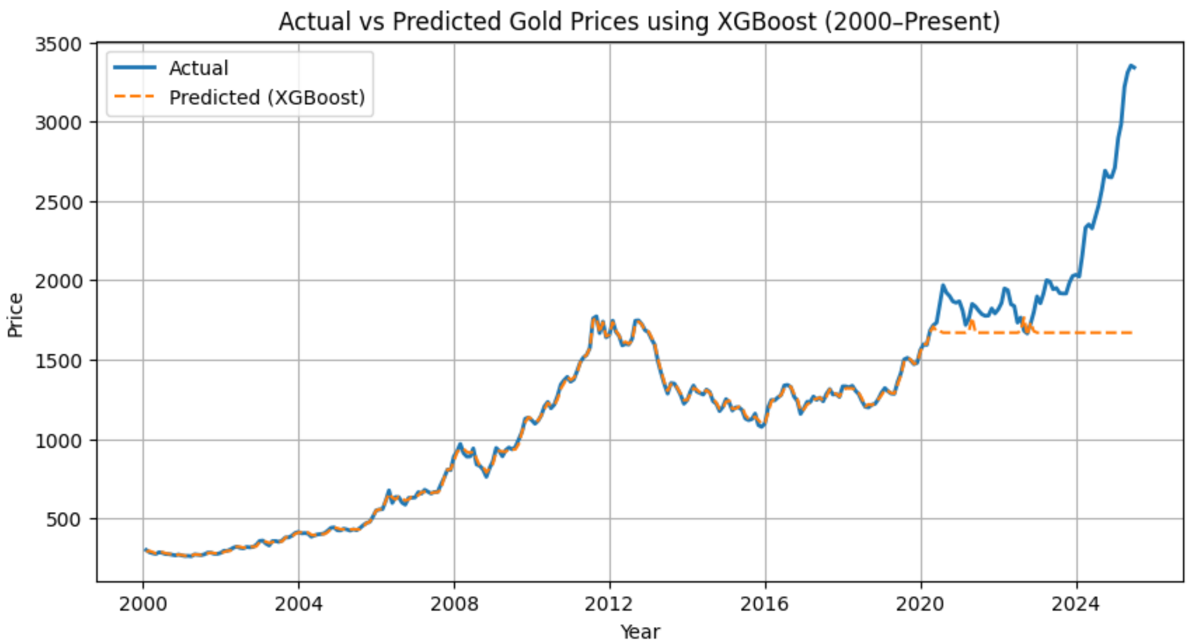
This graph compares the actual gold prices with the prices predicted by the Linear Regression model from 2000 to the present.
The predicted line shows a smooth upward trend, indicating that Linear Regression captures the long-term increasing pattern of gold prices. However, it fails to accurately follow sharp fluctuations and sudden price changes. This results in moderate prediction errors, as reflected by relatively higher RMSE and MAE values.

2) **Actual vs Predicted Gold Prices (Random Forest)**



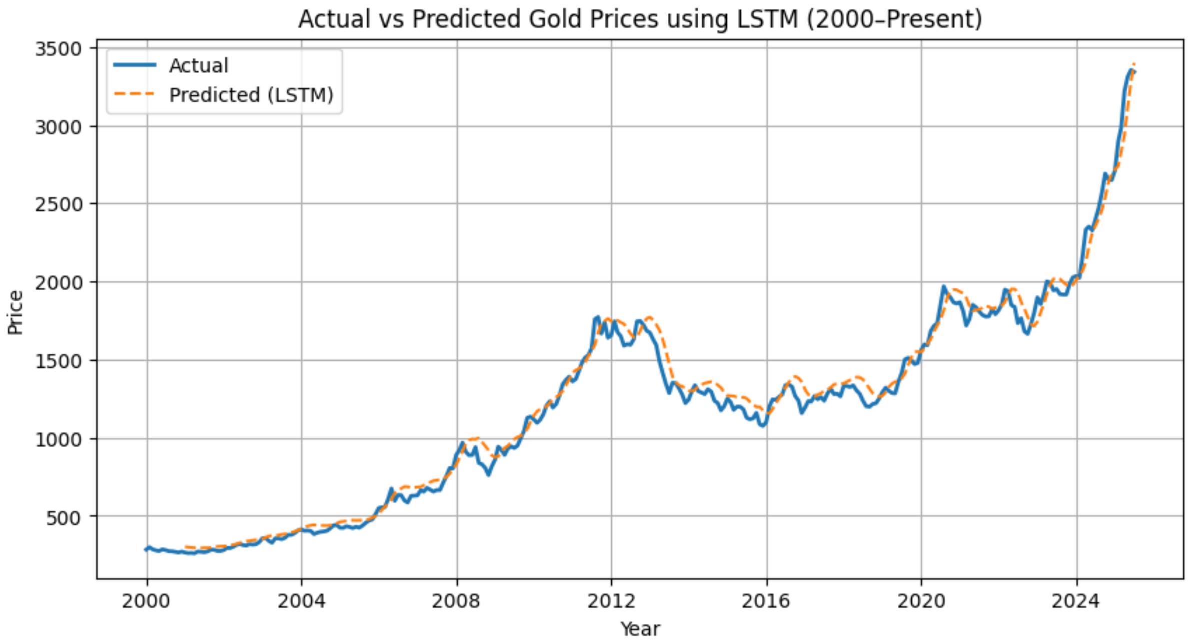
The Random Forest model predictions closely follow the actual gold prices during stable periods. However, the model shows limitations in capturing extreme price surges in recent years. The predicted values appear flattened after a certain point, indicating underfitting for highly volatile market conditions. This leads to higher error values compared to Linear Regression.

3) **Actual vs Predicted Gold Prices (XGBoost)**



The XGBoost model demonstrates similar behavior to Random Forest, effectively capturing historical trends but struggling with recent sharp upward movements in gold prices. Although it models non-linear relationships better than Linear Regression, prediction errors remain relatively high, indicating reduced forecasting accuracy during volatile periods.

4) **Actual vs Predicted Gold Prices (LSTM)**



The LSTM model shows a strong alignment between actual and predicted gold prices throughout the entire time period. It effectively captures both long-term trends and short-term fluctuations due to its ability to model sequential and temporal dependencies in time-series data. This results in significantly lower RMSE, MAE, and MAPE values, indicating superior forecasting performance.

Table : Performance Metrics Comparison of Different Models for Gold Price Prediction

Model	Category	RMSE	MAE	MAPE (%)	Interpretation
Linear Regression	Machine Learning	451.46	280.03	11.10	Captures overall trend but fails during sudden volatility.
Random Forest	Machine Learning	595.73	401.52	16.28	Learns historical patterns but underestimates extreme changes.
XGBoost	Machine Learning	616.04	426.93	17.50	Handles non-linearity but lacks sequential memory.
LSTM	Deep Learning	98.08	78.36	3.72	Accurately models long-term dependencies and volatility.

Lower values of RMSE, MAE, and MAPE indicate better forecasting accuracy.

Among all the models evaluated, **LSTM outperforms Linear Regression, Random Forest, and XGBoost** in forecasting gold prices.
The LSTM model achieves the **lowest RMSE (98.08), MAE (78.36), and MAPE (3.72%)**, demonstrating high prediction accuracy and robustness.

Traditional machine learning models such as Linear Regression, Random Forest, and XGBoost are limited in capturing complex temporal dependencies present in long-term financial time-series data. In contrast, LSTM, being a deep learning model designed for sequential data, successfully learns both historical patterns and recent market volatility.

Therefore, LSTM is concluded to be the best-performing model for gold price prediction in this study.

7. Conclusion

This study looked at how major global events affect gold prices using machine learning and deep learning models. Data from 2000 to 2025 shows that gold acts as a safe-haven asset, with prices rising during global crises like the Global Financial Crisis, COVID-19, and geopolitical tensions.

Linear Regression can capture general trends but cannot handle sudden spikes or complex patterns in prices.

LSTM models perform the best as they can learn both short-term changes and long-term trends in gold prices. Overall, machine learning and deep learning models give more accurate predictions than simple linear models.

Gold remains a reliable hedge during times of economic and geopolitical uncertainty.

8. Future Scope

Future studies can improve gold price prediction in several ways:

1. **Sentiment Analysis** – Use news, financial reports, and social media to understand investor reactions.
2. **Real-Time Data** – Use daily or real-time prices to predict short-term changes and react quickly to events.
3. **Hybrid Models** – Combine Linear Regression with machine learning or deep learning to capture both simple trends and complex patterns.
4. **Event and Risk Scores** – Include event severity or geopolitical risk indices to measure impact more accurately.
5. **Cross-Market Analysis** – Use data from currency markets, stocks, and global indicators for a better understanding of gold price movements.

9. References

1. Hussain, M., & Rehman, S. (2019). Inflation, currency depreciation, and gold price interactions. *Journal of Monetary Economics Studies*, 7(1), 88–102.
2. Jeon, S., & Kim, Y. (2020). Global uncertainty and gold price behavior during COVID-19. *Journal of Economic Studies*, 47(6), 1120–1135.
3. Chen, L., & Wu, J. (2020). Investor fear and gold price movements: A VIX-based analysis. *Journal of Behavioral Finance*, 21(4), 350–362.
4. Alqahtani, M. (2021). Geopolitical risk and gold price volatility. *Journal of International Financial Studies*, 9(4), 203–217.
5. Smith, J., & Carter, R. (2022). War, sanctions, and gold prices: Evidence from the Russia–Ukraine conflict. *Journal of Commodity Markets*, 30, 100–211.
6. Sharma, K. (2021). Effects of COVID-19 on global commodity markets: Evidence from gold prices. *Finance Research Letters*, 39, 101435.
7. Dahiya, A., & Verma, R. (2018). Impact of monetary policy announcements on gold prices. *Journal of Economic Policy Studies*, 5(2), 89–103.
8. Kumar, N., & Tiwari, S. (2020). Comparative analysis of machine learning algorithms for gold price prediction. *Journal of Intelligent Systems*, 29(4), 595–608.
9. Sharma, M., & Singh, H. (2021). XGBoost and ensemble methods for financial time-series forecasting: A case of gold prices. *Financial AI Journal*, 2(1), 33–49.
10. Mani, P., & Rajan, J. (2020). Deep learning models (LSTM and GRU) for gold price forecasting. *International Journal of Data Science*, 7(2), 143–160.
11. Das, A., & Panigrahi, A. (2021). Evaluation of machine learning techniques for commodity price prediction. *Journal of Computational Finance*, 24(1), 76–94.
12. Martin, L., & Lopez, A. (2022). Event-driven machine learning models for gold price forecasting. *Expert Systems with Applications*, 198, 116245.
13. Ahmed, I., & Farooq, M. (2021). Sentiment-driven gold price prediction using news analytics and machine learning. *Journal of Financial Data Science*, 3(2), 95–110.
14. Goyal, R., Sharma, V., & Singh, P. (2020). Hybrid ARIMA-machine learning models for gold price prediction. *International Journal of Forecasting*, 36(4), 1280–1293.
15. Patel, K., Shah, H., & Desai, N. (2021). Comparing ARIMA, LSTM, and hybrid models for gold price forecasting. *Journal of Data Science and Analytics*, 5(3), 211–225.
16. Wei, L., & Huang, T. (2022). Event-based gold price prediction using NLP and machine learning techniques. *Journal of Information Systems and Data Mining*, 4(2), 67–84.
17. Singh, A., & Rao, V. (2021). Analyzing the effect of global events on gold prices using text mining and machine learning. *Journal of Economic Computation*, 12(3), 190–205.
18. Baur, D. G., & McDermott, T. K. (2010). Is gold a safe haven? International evidence. *Journal of Banking and Finance*, 34(8), 1886–1898.

19. Beckmann, J., Berger, T., & Czudaj, R. (2015). Does gold act as a hedge or a safe haven for stocks? *Economic Modelling*, 48, 184–194.

20. Joy, M. (2011). Gold and the US dollar: Hedge or haven? *Finance Research Letters*, 8(3), 120–131.

21. Smales, L. A. (2016). Investor attention and safe-haven assets. *Finance Research Letters*, 17, 70–77.

22. Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194–1225.

23. Aysan, A. F., Demir, E., Gozgor, G., & Lau, C. K. M. (2019). Effects of geopolitical risks on gold prices. *Journal of International Financial Markets, Institutions and Money*, 58, 253–267.

24. Baur, D. G., & Lucey, B. M. (2010). Is gold a hedge or a safe haven? *Financial Review*, 45(2), 217–229.

25. Goodell, J. W. (2020). COVID-19 and finance: Agendas for future research. *Finance Research Letters*, 35, 101512.

26. Boubaker, S., Goodell, J. W., Pandey, D. K., & Kumari, V. (2020). Heterogeneous impacts of wars on global financial markets. *Journal of Financial Stability*, 50, 100784.

27. Bernanke, B. S. (2015). The federal reserve and the financial crisis. *Princeton University Press*, Princeton.

28. Atsalakis, G. S. (2016). Using computational intelligence techniques for financial forecasting. *Neural Computing and Applications*, 27(4), 1169–1181.

29. Chong, J., & Lin, M. (2019). Estimating gold prices using deep learning models. *Neural Computing and Applications*, 31(11), 7221–7234.

30. Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.

DataHub. (n.d.). Monthly gold prices [Dataset]. , from https://datahub.io/core/gold-prices/r/monthly.csv